WORKEDOUT EXAMPLES:

W.E-1: Let M be a 3×3 matrix satisfying

$$M\begin{bmatrix}0\\1\\0\end{bmatrix}=\begin{bmatrix}-1\\2\\3\end{bmatrix}, M\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}1\\1\\-1\end{bmatrix} \text{ and } M\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}0\\0\\12\end{bmatrix}$$

then the sum of the diagonal entries of M is (ADV-2011)

Sol:(9)

Let $M=\begin{bmatrix}a_1&a_2&a_3\\b_1&b_2&b_3\\c_1&c_2&c_3\end{bmatrix}$

$$\therefore M\begin{bmatrix}0\\1\\0\end{bmatrix}=\begin{bmatrix}-1\\2\\3\end{bmatrix}, M\begin{bmatrix}1\\-1\\0\end{bmatrix}=\begin{bmatrix}1\\1\\-1\end{bmatrix}, M\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}0\\0\\12\end{bmatrix}$$

$$\Rightarrow\begin{bmatrix}a_2\\b_2\\c_2\end{bmatrix}=\begin{bmatrix}-1\\2\\3\end{bmatrix}, \begin{bmatrix}a_1-a_2\\b_1-b_2\\c_1-c_2\end{bmatrix}=\begin{bmatrix}1\\1\\-1\end{bmatrix},$$

$$\begin{bmatrix}a_1+a_2+a_3\\b_1+b_2+b_3\\c_1+c_2+c_3\end{bmatrix}=\begin{bmatrix}0\\0\\12\end{bmatrix}$$

$\Rightarrow a_2=-1, b_2=2, c_2=3, a_1-a_2=1$

$b_1-b_2=1, c_1-c_2=-1$

$a_1+a_2+a_3=0, b_1+b_2+b_3=0, c_1+c_2+c_3=12$

$\therefore a_1=0, b_2=2, c_3=7$

$\Rightarrow$ Sum of diagonal elements= 0+2+7= 9

W.E-2: Let ω be a complex cube root of unity with $\omega\neq1$ and $P=\left[P_{ij}\right]$ be a $n\times n$ matrix with $P_{ij}=\omega^{i+j}$. Then $P^2\neq0$, when n=

A)57 B)55 C)58 D)56 (ADV-2014)

Sol:(B,C,D)

Here $P=\left[p_{ij}\right]_{m\times n}$ with $p_{ij}=\omega^{i+j}$

When n=1,

$P=\left[p_{ij}\right]_{1\times1}=\left[\omega^2\right]\Rightarrow P^2=\left[\omega^4\right]\neq0$

When n=2,

$$P=\left[p_{ij}\right]_{2\times2}=\begin{bmatrix}p_{11}&p_{12}\\p_{21}&p_{22}\end{bmatrix}=\begin{bmatrix}\omega^2&\omega^3\\\omega^3&\omega^4\end{bmatrix}$$

$$=\begin{bmatrix}\omega^2&1\\1&\omega\end{bmatrix}$$

$$P^2=\begin{bmatrix}\omega^2&1\\1&\omega\end{bmatrix}\begin{bmatrix}\omega^2&1\\1&\omega\end{bmatrix}$$

$$P^2=\begin{bmatrix}\omega^4+1&\omega^2+\omega\\\omega^2+\omega&1+\omega^2\end{bmatrix}\neq0$$

When n= 3

$$P=\left[p_{ij}\right]_{3\times3}=\begin{bmatrix}\omega^2&\omega^3&\omega^4\\\omega^3&\omega^4&\omega^5\\\omega^4&\omega^5&\omega^6\end{bmatrix}=\begin{bmatrix}\omega^2&1&\omega\\1&\omega&\omega^2\\\omega&\omega^2&1\end{bmatrix}$$

$$P^2=\begin{bmatrix}\omega^2&1&\omega\\1&\omega&\omega^2\\\omega&\omega^2&1\end{bmatrix}\begin{bmatrix}\omega^2&1&\omega\\1&\omega&\omega^2\\\omega&\omega^2&1\end{bmatrix}=\begin{bmatrix}0&0&0\\0&0&0\\0&0&0\end{bmatrix}=0$$

$\therefore P^2=0$, when n is multiple of 3.

$P^2\neq0$, when n is not a multiple of 3.

$\Rightarrow n=57$ is not possible

$\therefore n=55,58,56$ are possible.

W.E-3: Let M and N be two 3×3 matrices such that MN=NM, further,if $M\neq N^2$ and $M^2=N^4$ then

A)determinant of $\left(M^2+MN^2\right)$ is 0

B)There is a 3×3 non-zero matrix U such that $\left(M^2+MN^2\right)U$ is zero matrix

C)determinant of $\left(M^2+MN^2\right)\geq1$

D)for a 3×3 matrix u, if $\left(M^2+MN^2\right)U$ equal to the zero matrix, then U is the zero matrix

(ADV-2014)

Ans: (A)

Sol:

(i) If A and B are two non-zero matrices and $AB=BA$, then $(A-B)(A+B)=A^2-B^2$.

(ii) The determinant of the product of the matrices is equal to product of their individual determinants

i.e $|AB| = |A||B|$

Given, $M^2 = N^4 \Rightarrow M^2 - N^4 = 0$

$\Rightarrow (M - N^2)(M + N^2) = 0$ (as MN=NM)

Also, $M \neq N^2 \Rightarrow M + N^2 = 0$

$\Rightarrow Det(M + N^2) = 0$

Also,

$Det(M^2 + MN^2) = (DetM)(Det(M + N^2))$

$= (DetM)(0) = 0$

As, $Det(M^2 + MN^2)U = 0$

W.E-4: If $A^2B = BA$ then $(AB)^{20} = A^{\lambda}B^{20}$ then find the value of λ.

Sol:($2^{20} - 1$)

$(AB)^2 = A(BA)B = A.A^2B.B = A^3B^2 = A^{2^2-1}B^2$

$(AB)^3 = (AB)^2 AB = A^3B^2AB = A^3B(BA)B$

$= A^3BA^2B.B = A^3(BA)(AB)B$

$= A^3A^2BABB = A^5(BA)B^2$

$= A^5.A^2B.B^2 = A^7B^3$

Similarly $(AB)^{20} = A^{2^{20}-1}B^{20}$

$\therefore \lambda = 2^{20} - 1$

W.E-5: If A,B,C,D are four rectangular matrices such that $A^T = BCD, B^T = CDA, C^T = DAB$ and $D^T = ABC$. If $S = ABCD$ then find period of S.

Sol: (2) $S = ABCD = A(BCD)$

$\Rightarrow S = AA^T$ ....(1)

$S^2 = (ABCD)(ABCD) = (ABC)(DAB)CD$

$\Rightarrow S^2 = D^TC^T.CD$ ....(2)

$S^3 = (ABCD)(ABCD)(ABCD)$

$= (ABC)(DAB)(CDA)(BCD) = D^TC^TB^TA^T$

$S^3 = (BCD)^T A = (A^T)^T A$

$\Rightarrow S^3 = AA^T$ ....(3)

From equation (1) and (3) we can say that $S = S^3$, hence period of S is 2.

W.E-6: If A is square matrix such that $A - \frac{1}{2}I$ and $A + \frac{1}{2}I$ are orthogonal. Then prove that A is skew symmetric and find matrix $4A^2 + 3I$

Sol:

$As\ A - \frac{1}{2}I$ and $A + \frac{1}{2}I$ are orthogonal.

$$\left(A - \frac{1}{2}I\right)\left(A - \frac{1}{2}I\right)^T = I$$

$$\Rightarrow AA^T - \frac{A}{2} - \frac{A^T}{2} + \frac{I}{4} = I$$

$$\Rightarrow AA^T - \left(\frac{A + A^T}{2}\right) = \frac{3I}{4} \text{.............(1)}$$

and $\left(A + \frac{1}{2}I\right)\left(A + \frac{1}{2}I\right)^T = I$

$$\Rightarrow AA^T + \left(\frac{A + A^T}{2}\right) = \frac{3I}{4} \text{.............(2)}$$

From equation (1) and (2)

$$\frac{A + A^T}{2} = -\left(\frac{A + A^T}{2}\right) \Rightarrow A + A^T = 0$$

$\therefore\ \therefore A^T = -A$

i.e matrix A is skew symmetric

From equation(1)

$$AA^T - \left(\frac{A + A^T}{2}\right) = \frac{3I}{4}$$

$$\Rightarrow A(-A) = \frac{3I}{4} \Rightarrow 4A^2 + 3I = 0$$

W.E-7: If B and C are two square matrices of order n and also $A = B + C$, $BC = CB$, $C^2 = 0$ then prove that for every natural number K,

$A^{K+1} = B^K[B + (K+1)C]$

Sol: Let K= 1,then

$A^2 = B[B + 2C] \Rightarrow A^2 = B^2 + 2BC$

$(B + C)^2 = B^2 + 2BC$

$B^2 + C^2 + BC + CB = B^2 + 2BC$

$B^2 + 2BC = B^2 + 2BC$ (As $C^2 = 0$ and $BC = CB$)

Hence, equation is true for K= 1

Let the result is true for K where K is any +ve integer.Now

$$A^{(K+1)+1} = A^{K+1}.A = B^K\left(B+\left(K+1\right)C\right)\left(B+C\right)$$

$$= B^k\left[B^2 + BC + \left(K+1\right)CB + \left(K+1\right)C^2\right]$$

$$= B^K\left[B^2 + \left(K+2\right)BC\right]$$

$$A^{K+2} = B^{K+1}\left[B+\left(K+2\right)\right]$$

So, result is true for K+1 Hence result is true for each and every value of K.

W.E-8: For 3×3 matrices M and N which of the following statement(s) is (are) not correct?

(A) $N^T MN$ is symmetric or skew symmetric, according as M is symmetric or skew symmetric

(B) MN-NM is skew symmetric for all symmetric matrices M and N

(C) MN is symmetric for all symmetric matrices M and N

(D) (adjM) (adjN) = adj(MN) for all invertible matrices M and N

(ADV-2013)

Sol: a) $\left(N^1MN\right)^1 = \left(MN\right)^1 N = N^1M^1N = N^1MN$ or $-N^1MN$ according as M is symmetric or skew symmetric. $\therefore$ Correct

b) $\left(MN-NM\right)^1 = \left(MN\right)^1 - \left(NM\right)^1$

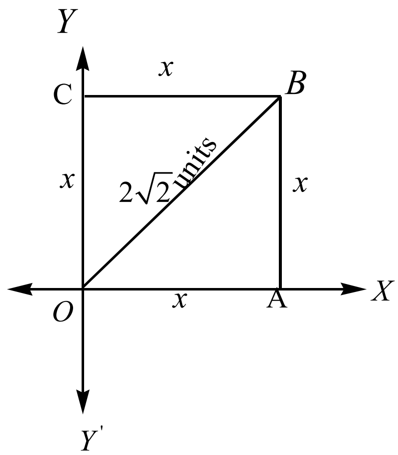
$= N^1M^1 - M^1N^1 = NM - MN$

skew symmetric $\therefore$ Correct

c) $\left(MN\right)^1 = N^1M^1 = NM \neq MN$ $\therefore$ Incorrect

d) $\left(\text{adjM}\right)\left(\text{adjN}\right) = \text{adj}\left(\text{MN}\right)$ $\therefore$ Incorrect

W.E-9: Write down 2×2 matrix A which corresponds to a counterclockwise rotation of 60^0 about the origin. In the diagram the square $OABC$ has its diagonal OB of $2\sqrt{2}$ units in length. The square is rotated counterclock wise about O through 60^0. Find the coordinates of the vertices of the square after rotating.



Sol: The matrix describes a rotation through an angle 60^0 in counterclock wise direction is

$$\begin{bmatrix}\cos 60^0 & -\sin 60^0\\ \sin 60^0 & \cos 60^0\end{bmatrix} = \begin{bmatrix}1/2 & -\sqrt{3}/2\\ \sqrt{3}/2 & 1/2\end{bmatrix}$$

since each side of the square be x then

$$x^2 + x^2 = \left(2\sqrt{2}\right)^2 \Rightarrow 2x^2 = 8 \Rightarrow x^2 = 4$$

$\therefore\ x = 2$ units.

Therefore, the co-ordinates of the vertices O, A, B, C are $\left(0,0\right),\left(2,0\right),\left(2,2\right),\left(0,2\right)$ respectively. Let after rotation A map into A', B map into B', C map into C', but the O map into itself.

if Co-ordinates of A', B' & C' are $\left(x_1, y_1\right),\left(x_2, y_2\right) and \left(x_3, y_3\right)$ respectively.

$$\therefore \begin{bmatrix}x_1\\ y_1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1 & -\sqrt{3}\\ \sqrt{3} & 1\end{bmatrix}\begin{bmatrix}2\\ 0\end{bmatrix} = \begin{bmatrix}1\\ \sqrt{3}\end{bmatrix}$$

$\therefore x_1 = 1, y_1 = \sqrt{3}$

$\Rightarrow A\left(2,0\right) \to A'\left(1,\sqrt{3}\right)$ and

$$\begin{bmatrix}x_2\\ y_2\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1 & -\sqrt{3}\\ \sqrt{3} & 1\end{bmatrix}\begin{bmatrix}2\\ 2\end{bmatrix} = \frac{1}{2}\begin{bmatrix}2-2\sqrt{3}\\ 2\sqrt{3}+2\end{bmatrix} = \begin{bmatrix}1-\sqrt{3}\\ \sqrt{3}+1\end{bmatrix}$$

$\therefore x_2 = 1-\sqrt{3}, y_2 = \sqrt{3}+1$

$\Rightarrow B\left(2,2\right) \to B'\left(1-\sqrt{3}, \sqrt{3}+1\right)$

$$\begin{bmatrix}x_3\\ y_3\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1 & -\sqrt{3}\\ \sqrt{3} & 1\end{bmatrix}\begin{bmatrix}0\\ 2\end{bmatrix} = \frac{1}{2}\begin{bmatrix}-2\sqrt{3}\\ 2\end{bmatrix} = \begin{bmatrix}-\sqrt{3}\\ 1\end{bmatrix}$$

$\therefore x_3 = -\sqrt{3}, y_3 = 1$

$\Rightarrow C(0,2) \to C''\left(-\sqrt{3},1\right)$

W.E-10: Show that the system of equations $3x - y + 4z = 3$ **,** $x + 2y - 3z = -2$ **and** $6x + 5y + \lambda z = -3$ **has at least one solution for any real number** $\lambda \neq -5$**. Find the set of solutions, if** $\lambda = -5$. **(ADV - 1983)**

Sol: For $\lambda \neq -5$ the system has atleast one solution

for $\lambda = -5$ the system has infinitly many solutions

$$x = \frac{4-5k}{7}, y = \frac{13k-9}{7}, z = k\,(k \in R)$$

W.E-11: Let $\alpha_1, \alpha_2, \beta_1, \beta_2$ **be the roots of** $ax^2 + bx + c = 0$ **and** $px^2 + qx + r = 0$ **respectively. If the system of equations** $\alpha_1 y + \alpha_2 z = 0$ **and** $\beta_1 y + \beta_2 z = 0$ **has a non-trivial solution, then prove that** $\frac{b^2}{q^2} = \frac{ac}{pr}$**.** **(IIT-1987)**

Sol:

Since, α_1, α_2 are the roots of $ax^2 + bx + c = 0$

$$\Rightarrow \alpha_1 + \alpha_2 = -\frac{b}{a} \text{ and } \Rightarrow \alpha_1\alpha_2 = \frac{c}{a} \text{ ------(1)}$$

Also, β_1, β_2 are the roots of $px^2 + qx + r = 0$

$$\Rightarrow \beta_1 + \beta_2 = -\frac{q}{p} \text{ and } \beta_1\beta_2 = \frac{r}{p} \text{ ------(2)}$$

Given system of equations

$\alpha_1 y + \alpha_2 z = 0$

and $\beta_1 y + \beta_2 z = 0$, has non - trivial solution.

$$\therefore \begin{vmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{vmatrix} = 0 \Rightarrow \frac{\alpha_1}{\alpha_2} = \frac{\beta_1}{\beta_2}$$

Applying componendo - dividendo

$$\frac{\alpha_1 + \alpha_2}{\alpha_1 - \alpha_2} = \frac{\beta_1 + \beta_2}{\beta_1 - \beta_2}$$

$$\Rightarrow (\alpha_1 + \alpha_2)(\beta_1 - \beta_2) = (\alpha_1 - \alpha_2)(\beta_1 + \beta_2)$$

$$\Rightarrow (\alpha_1 + \alpha_2)^2\{(\beta_1 + \beta_2)^2 - 4\beta_1\beta_2\}$$

$$= (\beta_1 + \beta_2)^2\left\{(\alpha_1 + \alpha_2)^2 - 4\alpha_1\alpha_2\right\}$$

From Eqs. (1) and (2), we get

$$\frac{b^2}{a^2}\left(\frac{q^2}{p^2} - \frac{4r}{p}\right) = \frac{q^2}{p^2}\left(\frac{b^2}{a^2} - \frac{4c}{a}\right)$$

$$\Rightarrow \frac{b^2q^2}{a^2p^2} - \frac{4b^2r}{a^2p} = \frac{b^2q^2}{a^2p^2} - \frac{4q^2c}{ap^2}$$

$$\Rightarrow \frac{b^2r}{a} = \frac{q^2c}{p} \Rightarrow \frac{b^2}{q^2} = \frac{ac}{pr}$$

W.E-12: The system of linear equations, x + y + z = 6, x + 2y + 3z = 14 and 2x + 5y + λ **z =** μ **have**

(A) Infinitely many solutions when λ **= 8 and** μ **= 36**

(B) Unique solution when $\lambda \neq 8$

(C) No solution when λ **= 8 and** $\mu \neq 36$

(D) No solution when λ **= 8 and** $\mu = 36$

Sol: (A,B,C) x + y + z = 6

x + 2y + 3z = 14

2x + 5y + λ z = μ

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 5 & \lambda \end{vmatrix}, \quad D_x = \begin{vmatrix} 6 & 1 & 1 \\ 14 & 2 & 3 \\ \mu & 5 & \lambda \end{vmatrix}$$

$$D_y = \begin{vmatrix} 1 & 6 & 1 \\ 1 & 14 & 3 \\ 2 & \mu & \lambda \end{vmatrix}, \quad D_z = \begin{vmatrix} 1 & 1 & 6 \\ 1 & 2 & 14 \\ 2 & 5 & \mu \end{vmatrix}$$

For unique solution $D \neq 0$ see λ for D non zero.

For infinitely many solution $D = D_x = D_y = D_z = 0$

Solve for value of λ and μ.

W.E-13: If $2x + py + 6z = 8, x + 2y + qz = 5,$ $x + y + 3z = 4$**. Then for which the planes (i) have no common point (ii) have only one common point (iii)intersect in a line**

Sol: Here $\Delta = \begin{vmatrix} 2 & p & 6 \\ 1 & 2 & q \\ 1 & 1 & 3 \end{vmatrix} = (p-2)(q-3)$

$$\Rightarrow \Delta_1 = \begin{vmatrix} 8 & p & 6 \\ 5 & 2 & q \\ 4 & 1 & 3 \end{vmatrix} = (4q-15)(p-2) \text{ and}$$

$$\Rightarrow \Delta_2 = \begin{vmatrix} 2 & 8 & 6 \\ 1 & 5 & q \\ 1 & 4 & 3 \end{vmatrix} = 0$$

$$\Rightarrow \Delta_3 = \begin{vmatrix} 2 & p & 8 \\ 1 & 2 & 5 \\ 1 & 1 & 4 \end{vmatrix} = p-2$$

Case 1:

$\Delta \neq 0$ i.e., $p \neq 2$ and $q \neq 3$, then given system of equation has unique solution.

Case 2:

When $\Delta = 0$ i.e., $p = 2$ or $q = 3$

$\Rightarrow$ When $p = 2, \Delta = 0, \Delta_1 = 0, \Delta_2 = 0, \Delta_3 = 0$

Given system of equations has no solution,

Concluding above, we can say that

(i) no solution when $p \neq 2, q = 3$.

(ii) unique solution when $p \neq 2$ and $q \neq 3$

(iii)infinetely many solutions when $p = 2$

W.E-14: Show that the system $ax + y + z = a$, $x + by + z = b$ and $x + y + cz = c$ is inconsistent if $abc - a - b - c + 2 = 0$ and a,b,c are different from unity.

Sol: We have

$$\Delta = \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = abc - a - b - c + 2$$

$$\Delta_x = \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = abc - a + b + c - 2bc$$

$\Delta_y = abc - b + a + c - 2ac$

and $\Delta_z = abc - c + a + b - 2ab$

But for a system to be inconsistent, it is sufficient that $\Delta = 0$ and least one of $\Delta_x, \Delta_y, \Delta_z$ is non-zero. Rewriting Δ_x as

$\Delta_x = abc - a - b - c + 2 + 2b + 2c - 2bc - 2$

$= 0 + 2b + 2c - 2bc - 2 \qquad (\because \Delta = 0)$

$= -2(1-b)(1-c)$

Similarly $\Delta_y = -2(1-c)(1-a)$

$\Delta_z = -2(1-a)(a-c)$

Now if a,b,c are different from unity $\Delta_x, \Delta_y, \Delta_z$ cannot vanish .Hence the result follows.

LEVEL -V

SINGLE ANSWER QUESTIONS

1. $A = \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$ **and** $MA = A^{2m}, m \in N$ **for some matrix M, then which one of the following is correct?**

A) $M = \begin{bmatrix} a^{2m} & b^{2m} \\ b^{2m} & -a^{2m} \end{bmatrix}$

B) $M = (a^2 + b^2)^m \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

C) $M = (a^m + b^m) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

D) $M = (a^2 + b^2)^{m-1} \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$

2. If $A = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}, I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $A^2 = 8A + \alpha I$, then, the value of α is

(A) 7 (B) 8 (C) -7 (D) -8

3. If $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ **and** $B = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$ **then** $(A+B)^2 =$

(A) A (B) B (C) I (D) $A^2 + B^2$

4. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ **and** $B = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ $a,b \in N$ **then**

A) there exists exactly one B such that AB=BA

B) There exists infinitely many B's such that AB=BA

C) there cannot exist any B such that AB=BA

D) there exists more than one but finite number of B's such that AB=BA

5. If $A=\begin{bmatrix}1 & 2\\ -1 & 3\end{bmatrix}$ and if $A^6=KA-205I$ then

(A) $K=11$ (B) $K=22$

(C) $K=33$ (D) $K=44$

6. If $A=\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 0\\ 5 & 1\end{bmatrix}$ then the value of α for which $A^2=B$ is [IIT-2003]

(A) 1 (B) -1

(C) 4 (D) no real values.

7. Let A and B be two square matrices of the same order such that $AB=BA, A^m=O$, $B^n=O$ for some integers m and n, such that greatest common divisor of m, n is 1. Then the least positive integer 'r' such that $(A+B)^r=O$ is

A) $m+n$ B) $m-n$

C) GCD $\{m,n\}$ D) LCM $\{m,n\}$

8. Let $B=A^3-2A^2+3A-I$ where I is a unit matrix and A = $\begin{bmatrix}1 & 3 & 2\\ 2 & 0 & 3\\ 1 & -1 & 1\end{bmatrix}$ then the transpose of matrix B is equal to

(A) $\begin{bmatrix}8 & 14 & 7\\ 21 & 1 & -7\\ 14 & 21 & 8\end{bmatrix}$ (B) $\begin{bmatrix}2 & 21 & 14\\ 14 & 1 & 21\\ 7 & -7 & 8\end{bmatrix}$

(C) $\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$ (D) $\begin{bmatrix}3 & 1 & 0\\ 1 & 1 & 0\\ 3 & 1 & 0\end{bmatrix}$

9. Let A is a 3×3 matrix and $A=\left[a_{ij}\right]_{3\times3}$. If for every column matrix X, if $X^TAX=O$ and $a_{23}=-2009$ then $a_{32}=$

(A) 2009 (B) -2009 (C) 0 (D) 2008

10. If $A=\begin{bmatrix}\cos\theta & \sin\theta\\ \sin\theta & -\cos\theta\end{bmatrix}, B=\begin{bmatrix}1 & 0\\ -1 & 1\end{bmatrix}$, $C=ABA^T$ then A^TC^nA equals to $(n\in N)$

A) $\begin{bmatrix}-n & 1\\ 1 & 0\end{bmatrix}$ B) $\begin{bmatrix}1 & -n\\ 0 & 1\end{bmatrix}$

C) $\begin{bmatrix}0 & 1\\ 1 & -n\end{bmatrix}$ D) $\begin{bmatrix}1 & 0\\ -n & 1\end{bmatrix}$

11. Let $\omega\neq1$ be be a cube root of unity and S be the set of all non-singular matrices of the form $\begin{bmatrix}1 & a & b\\ \omega & 1 & c\\ \omega^2 & \omega & 1\end{bmatrix}$, where each of a,b and c is either ω or ω^2. Then the number of distinct matrices in the set S is (IIT - 2011)

A) 2 B) 6 C) 4 D) 8

12. If α,β,γ are the roots of the equation $x^3-12x^2+47x-60=0$ and $P(\alpha,\beta,\gamma)$ lines in the plane $8x+4y+3z=20$ and A= $\begin{bmatrix}\alpha & 3 & 2\\ 1 & \beta & 4\\ 2 & 2 & \gamma\end{bmatrix}$ then $A(adjA)=$

A) $\begin{bmatrix}64 & 0 & 0\\ 0 & 64 & 0\\ 0 & 0 & 64\end{bmatrix}$ B) $\begin{bmatrix}34 & 0 & 0\\ 0 & 34 & 0\\ 0 & 0 & 34\end{bmatrix}$

C) $\begin{bmatrix}68 & 0 & 0\\ 0 & 68 & 0\\ 0 & 0 & 68\end{bmatrix}$ D) $\begin{bmatrix}20 & 0 & 0\\ 0 & 20 & 0\\ 0 & 0 & 20\end{bmatrix}$

13. The inverse of the matrix A = $\begin{bmatrix}1 & 0 & 0\\ a & 1 & 0\\ b & c & 1\end{bmatrix}$ is

(A) $\begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ ac-b & -c & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 & 0 \\ -a & 0 & 0 \\ b & -c & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ ac & b & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix}$

14. **If** $A=\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & a & 1 \end{bmatrix}$ **and** $A^{-1}=\begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & c \\ 5/2 & -3/2 & 1/2 \end{bmatrix}$ **then**

(A) $a=2, c=1/2$ (B) $a=1, c=-1$
(C) $a=-1, c=1$ (D) $a=1/2, c=1/2$

15. **If** $adj\ B = A$ **and** $|P|=|Q|=1$ **then** $adj\left(Q^{-1}BP^{-1}\right)=$

(A) APQ (B) PAQ (C) B (D) A

16. **Let f(x) =** $\frac{1+x}{1-x}$. **If A is matrix for which** $A^3=0$ **then** $f(A)$ **is**

A) $I+A+A^2$ **B)** I+2A+2A^2
C) $I-A-A^2$ **D) 0**

17. **Let P be a non-singular matrix and** $I+P+P^2+.......+P^n=O$ **then** P^{-1} **is**

a) P^n b) P c) P^{n-1} d) I

18. **If** $F(x)=\begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$,

$G(y)=\begin{bmatrix} \cos y & 0 & \sin y \\ 0 & 1 & 0 \\ -\sin y & 0 & \cos y \end{bmatrix}$ **then**

$Adj(F(x)G(y))=$

(A) F (x) G(-y) (B) $F^{-1}(x)G^{-1}(y)$

(C) $G^{-1}(-y)F^{-1}(-x)$ (D) $G(-y)F(-x)$

19. **Let** $\alpha=\pi/5$ **and** $A=\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$ **then** $B=A^{11}+A^{12}+A^{13}+A^{14}$ **is**

(A) singular (B) non-singular
(C) symmetric (D) Idempotent

20. **The matrix** $\begin{bmatrix} \frac{b^2-a^2}{a^2+b^2} & \frac{-2ab}{a^2+b^2} \\ \frac{-2ab}{a^2+b^2} & \frac{a^2-b^2}{a^2+b^2} \end{bmatrix}$ **is**

A) Idempotent matrix B) Nilpotent matrix
C) Orthogonal matrix D) Unit matrix

21. **If** $i=\sqrt{-1}$, $a=\frac{1+\sqrt{5}}{2}$, $b=\frac{1-\sqrt{5}}{2}$ **then which of the following matrix is idempotent**

(A) $\begin{bmatrix} a & i \\ i & -b \end{bmatrix}$ (B) $\begin{bmatrix} b & i \\ i & a \end{bmatrix}$

(C) $\begin{bmatrix} a & i \\ i & b \end{bmatrix}$ (D) $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$

22. **P is an orthogonal matrix and A is a periodic matrix with period 4,** $Q=PAP^T$ **then** $X=P^TQ^{2021}P$ **is equal to**

A) A B) A^2 C) A^3 D) A^4

23. **The point of intersection of the planes**

$x+2\omega y+3\omega^2 z=(1+\omega)(1-\omega)$

$2x+3\omega y+\omega^2 z=\omega(\omega-1), 3x+\omega y+2\omega^2 z=\omega-1$

(where ω **is an imaginary cube root of unity,** $1+\omega+\omega^2=0, \omega^3=1$**) is**

A) (1, 1, 1) B) (1, –1, 1)
C) (1, –1, –1) D) (–1, –1, –1)

24. **The number of** 3×3 **matrices A whose entries are either 0 or 1 and for which the system** $A\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ **has exactly two distinct solutions is**

A) 0 B) 2^9-1 C) 168 D) 2

25. **For the equations** $x+2y+3z=1$, $2x+y+3z=2,\ 5x+5y+9z=4$
(A) there is only one solution
(B) there exist infinitely many solutions
(C) there is no solution
(D) the equations are inconsistent.

26. **The system of equations** $6x+5y+\lambda z=0$, $3x-y+4z=0, x+2y-3z=0$ **has**
(A) only a trivial solution for $\lambda \in R$
(B) exactly one nontrivial solution for some real λ
(C) infinite number of nontrivial solutions for one value of λ
(D) none of these.

27. **One of the values of k for which the planes** $kx+4y+z=0$, $4x+ky+2z=0$ **and** $2x+2y+z=0$ **intersect in a straight line**
A) 0 B) 1 C) 2 D) 3

28. **Let 'X' be the solution set of the equation** $A^x=I$ **where** $A=\begin{bmatrix}0&1&-1\\4&-3&4\\3&-3&4\end{bmatrix}$ **and I is the corresponding unit matrix and** $x\in N$**.Then the minimum value of** $\sum(\cos^x\theta+\sin^x\theta),\theta\in R$ **is**
A) 2 B) 4 C) 6 D) 8

29. **If P is a** 3×3 **matrrix such that** $P^T=2P+I$ **where** P^T **is the transpose of P and I is** 3×3 **identity matrix then there exists a column matrix** $X=\begin{bmatrix}x\\y\\z\end{bmatrix}\neq\begin{bmatrix}0\\0\\0\end{bmatrix}$ **such that (IIT - 2012)**
A) $PX=\begin{bmatrix}0\\0\\0\end{bmatrix}$ B) $PX=X$
C) $PX=2X$ D) $PX=-X$

30. **If 2ax - 2y + 3z = 0, x + ay + 2z = 0 and 2x + az = 0 have a non-trivial solution then**
(A) a = 2 (B) a = 1 (C) a = 0 (D) a = -1

31. **If the equation 2x + 3y + 1 = 0, 2x + y - 1 = 0 and ax + 2y - b = 0 are consistent then**
(A) a - b = 2 (B) a + b + 1 = 0
(C) a + b = 3 (D) a - b - 8 = 0

MULTI ANSWER QUESTIONS

32. **If** $A=\begin{bmatrix}\alpha_1&\alpha_2\\\beta_1&\beta_2\end{bmatrix}$ **(where** $\alpha_2\neq\beta_1$ **and** $\alpha_1,\alpha_2,\beta_1,\beta_2$ **are non-zero) satisfies the equation** $x^2+k=0$ **then**
A) trace A = 0 B) $\alpha_1\beta_2<0$
C) det A = 0 D) det A = -1

33. **If** $A=\begin{bmatrix}1&0\\2&1\end{bmatrix}$ **and** $A^{2012}=\begin{bmatrix}a&b\\c&d\end{bmatrix}$ **then which of the following is(are) correct**
a) a=d b) a+b+c+d = 4026
c) $a^2+b^2+d^2=2$ d) b = 2012

34. **If A is** 3×3 **matrix whose** $(i,j)^{th}$ **element is given by** $a_{ij}=\begin{cases}2 & if\ i=j\\-1 & if\ |i-j|=1\\0 & else\ where\end{cases}$ **then**
A) A is symmetric B) Trace A = 6
C) det A is a perfect square
D) A^{-1} is skew symmetric

35. **Given that** $A=\begin{bmatrix}5&7&3\\1&5&2\\3&2&1\end{bmatrix}$ **and** $A^3-kA^2+\ell A=I$ **then**
A) K = 11 B) $\ell=15$ C) k = -11 D) $\ell=10$

36. **About a square matrix the following statements given are observed**
1) Sum of the eigen values of A is the trace of A
2) The product of eigen values of A is equal to the determinant of A
3) All eigen values of A are non - zero if and only if A is non - singular
4) If A^{-1} **exists then the eigen values of** A^{-1} **are equal to the reciprocals of eigen values of A**

Which of the following will be correct w.r.t the above statements

A) 3,4 are true

B) 2,3 are true and 1 is false

C) 2,3,4 are true

D) 1,2,3,4 are true

37. **Let** $A=\begin{bmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{bmatrix}, B=\begin{bmatrix}\cos 2\beta & \sin 2\beta\\ \sin 2\beta & -\cos 2\beta\end{bmatrix}$

where $0<\beta<\dfrac{\pi}{2}$ **then which of the following is(are) true**

A) $(AB)^2=I$ B) $(AB)^{-1}=AB$ C) $BAB=A^{-1}$

D) The least positive value of α for which $BA^4B=A^{-1}$ is $\dfrac{2\pi}{3}$

38. **If** $A=\begin{bmatrix}0 & 0 & -1\\ 0 & -1 & 0\\ -1 & 0 & 0\end{bmatrix}$ **then**

A) Trace of A is -1 B) Trace of A^{2012} *is* 3

C) A is an involutary matrix

D) Trace of A^{2013} *is* 1

39. **If the matrix** $\begin{bmatrix}l_1 & m_1 & n_1\\ l_2 & m_2 & n_2\\ l_3 & m_3 & n_3\end{bmatrix}$ **is orthogonal then**

A) $(l_1,m_1,n_1),(l_2,m_2,n_2),(l_3,m_3,n_3)$ can be the Dc's of a line

B) The rays with Dc's (l_1,m_1,n_1), $(l_2,m_2,n_2),(l_3,m_3,n_3)$ are orthogonal

C) The rays are parallel

D) The rays lie on the same plane

40. **Which of the following statements are false**

A) If A and B are square matrices of same order such that ABAB=0 then it follows that BABA = 0

B) Let A and B be different nxn matrices with real entries. If $A^3=B^3$ and $A^2B=B^2A$ then A^2+B^2 is invertible

C) If A is square singular and symmetic matrix then $\left(\left(A^{-1}\right)^{-1}\right)^{-1}$ is skew symmetric

D) The matrix of product of two invertible square matrices of same order is also invertible.

41. **If** $A=\begin{bmatrix}0 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{bmatrix}$ **then**

(A) Adj A is a zero matrix

(B) Adj $A=\begin{bmatrix}0 & 0 & -1\\ 0 & -1 & 0\\ -1 & 0 & 0\end{bmatrix}$

(C) A^{-1} = A (D) $A^2=I$

42. **P is a non-singular matrix and A,B are two matrices such that** $B=P^{-1}AP$ **then the true statements among the following are**

A) A is invertible iff B is invertible

B) $B^n=P^{-1}A^nP\ \forall n\in N$

C) $\forall\lambda\in R, B-\lambda I=P^{-1}(A-\lambda I)P$

D) A,B are both singular matrices

43. **If A and B are respectively a symmetric and a skew symmetric matrix such that AB = BA then**

A) $(A-B)^{-1}(A+B)$ is orthogonal matrix when $(A-B)$ is non- singular.

B) $(A+B)^{-1}(A-B)$ is orthogonal matrix when $(A+B)$ is non- singular.

C) $\det\left[(A-B)^{-1}(A+B)\right]=1$ and $\det\left[(A+B)^{-1}(A-B)\right]=1$

D) $\det\left[(A-B)^{-1}(A+B)\right]=-1$ and $\det\left[(A+B)^{-1}(A-B)\right]=1$

44. **Which of the following statement (s) is/are true about square matrix A of order n?**

A) $(-A)^{-1}$ **is equal to** $-A^{-1}$ **when n is odd only**

B) If $A^n=0,$ **then**

$I + A + A^2 ..... + A^{n-1} = (I-A)^{-1}$

C) If A is a skew symmetric matrix of odd order then its inverse does not exist

D) $\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$ **holds always**

45. **If the adjoint of a** 3×3 **matrix P is** $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$ **then the possible values of the determinant of P is (are)** **(IIT - 2012)**

A) -2 B) -1 C) 1 D) 2

46. **If A is symmetic and B is skew symmetric and A+B is non singular and also** $C = (A+B)^{-1}(A-B)$ **then**

A) $C^T(A+B)C = A+B$

B) $C^T(A-B)C = A-B$

C) $C^T AC = A$ D) $C^T AC = O$

47. **The matrix** $A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ **is**

A) Idempotent B) Involutary

C) $A^{-1} = A$ D) adj A = A

48. **If S is a real skew symmetric matrix then which of the following is true**

A) I - S is non-singular

B) $(I+S)(I-S)^{-1}$ is orthogonal

C) $(I+S)(I-S)^{-1}$ is non-singular

D) I+S is non- singular

49. **If a > b > c > 0 and system of equation ax + by + cz = 0 ,bx + cy + az = 0 and cx + ay + bz = 0 have a non-trivial solution then both roots of the equation at² + bt + c = 0 are**

(A) real (B) of opposite sign

(C) positive (D) complex

COMPREHENSION TYPE QUESTIONS

Paragraph - I

If $A_0 = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ **and**

$B_0 = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$, $B_n = adj(B_{n-1}), n \in N$

and I **is an identity matrix of order 3.** C_1, C_2, C_3 **represent the column matrix of** B_0 **as shown**

$C_1 = \begin{bmatrix} -4 \\ 1 \\ 4 \end{bmatrix}, C_2 = \begin{bmatrix} -3 \\ 0 \\ 4 \end{bmatrix}, C_3 = \begin{bmatrix} -3 \\ 1 \\ 3 \end{bmatrix}$ **then**

50. **det** $(A_0 + A_0^2 B_0^2 + A_0^3 + A_0^4 B_0^4 + ...$ **up to 10 terms) =**

A) 1000 B) -800 C) 0 D) -8000

51. $B_1 + B_2 + ... + B_{49} =$

A) B_0 B) 7 B_0 C) 49 B_0 D) 51 B_0

52. **For a variable matrix the equation** $A_0 X = C_1$ **will have**

A) unique solution B) No solution

C) Infinitely many solutions D) None

Paragraph - II

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ **satisfies** $A^n = A^{n-2} + A^2 - I$ **for** $n \geq 3$**. Further consider a matrix** $B_{3\times3}$ **with colums** B_1, B_2, B_3 **such that**

$A^{50}B_1 = \begin{bmatrix} 1 \\ 25 \\ 25 \end{bmatrix}, A^{50}B_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, A^{50}B_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

then

53. **Determinant of** $A^{50} =$

A) 0 B) 1 C) -1 D) 25

54. **Trace of** $A^{50} =$

A) 0 B) 1 C) 2 D) 3

55. **Sum of product of elements of B_1 with its corresponding cofactors in the adjoint matrix B is equal to**
A) Trace of B B) Det (2B)
C) Trace of A^2 D) Det (B^3)

Paragraph - III

For a given square matrix A, if there exists a matrix B such that $AB = BA = I$ then B is called inverse of A. Every non–singular square matrix possesses inverse and it exists

if $|A| \neq 0$, $A^{-1} = \frac{\text{adj}(A)}{\det(A)} \Rightarrow \text{adj}A = |A|(A^{-1})$.

56. **Let a matrix A** $= \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ **then it will satisfy the equation**
(A) $A^2 - 4A + I = O$ (B) $A^2 + 4A + I = O$
(C) $A^2 - 4A - 5I = O$ (D) $A^2 - 4A + 5I = O$

57. **Let a matrix A** $= \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ **then A^{-1} will be**
(A) $\begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$ (B) $\begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}$
(C) $\begin{bmatrix} -1 & 2 \\ 2 & -3 \end{bmatrix}$ (D) $\begin{bmatrix} -2 & 3 \\ 1 & -2 \end{bmatrix}$

58. **Let matrix A** $= \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$ **satisfies the equation** $A^2 + aA + bI = O$ **then the value of** $|a + b| =$
(A) 1 (B) 2 (C) 3 (D) 4

Paragraph - IV

Let
$$a_{11}x_1 + a_{12}x_2 + \ldots\ldots\ldots\ldots a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \ldots\ldots\ldots\ldots a_{2n}x_n = b_2$$
- - - - - - - - - - - - - - - - - -
- - - - - - - - - - - - - - - - - -
$$a_{n1}x_1 + a_{n2}x_2 + \ldots\ldots\ldots\ldots\ldots + a_{nn}x_n = b_n$$ **be a system of n linear equations in n unknowns. Then this can be written in the matrix form as AX=B Where**

A= $[a_{ij}]_{n\times n}, X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ . \\ . \\ . \\ x_n \end{pmatrix}, B = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ . \\ . \\ . \\ b_n \end{pmatrix}$ **Then**

(I) If |A| $\neq$ 0, the system is consistent, and has a unique solution given by $X = A^{-1}B$
(II) If |A| =0 and (adj A) B =0, then the system is consistent and has infinitely many solutions.
(III) If $|A|$ = 0 and (adj A) B $\neq$ 0, then the system is inconsistent.

59. **The system of equations** $2x - y + 3z = 1, x + y - 2z = 5, x + y + z = -1$ **has**
(A) a unique solution
(B) infinitely many solutions
(C) no solution
(D) finite number of solutions.

60. **Let** $2x - y + z = 4,\ x + 3y + 2z = 12,\ 3x + 2y + kz = 10$. **The value of k in the above system of equations so that system does not have a unique solution is**
(A) 2 (B) 3 (C) -1 (D) -2

61. **If** $x + y + z = 6,\ x + 2y + 3z = 10,\ x + 2y + \lambda z = \mu$ **the values of λ and μ for which the system has infinitely many solutions is**
(A) $\lambda = 3, \mu = 9$ (B) $\lambda = 3, \mu = 10$
(C) $\lambda = 2, \mu = 10$ (D) $\lambda = 10, \mu = 3$

Paragraph - V

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ **be a square matrix and**

C_1, C_2, C_3 be 3 column matrices satisfying $AC_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, AC_2 = \begin{bmatrix}2\\3\\0\end{bmatrix}$ and $AC_3 = \begin{bmatrix}2\\3\\1\end{bmatrix}$ of matrix B. If the matrix $C = \frac{1}{3}AB$ then

62. The value of sum of elements of B^{-1} is

A) -1 B) 0 C) 4 D) 2

63. The ratio of trace of matrix A to the det of matrix B is

A) 1 : 3 B) 2 :3 C) 1 : 1 D) 3 : 1

64. The value of $\sin^{-1}|A| + \cos^{-1}|C|$ is (where $|A|$ is determinant of A)

A) $\frac{\pi}{2}$ B) $\frac{\pi}{3}$ C) $\frac{\pi}{4}$ D) 1

Paragraph - VI

Consider the system of equations $\begin{bmatrix}3 & -2 & 1\\5 & -8 & 9\\2 & 1 & a\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}b\\3\\-1\end{bmatrix}$.

Now answer the following questions.

65. The values of a,b for which the system has no solution

A) $a = -3, b \in I$ B) $a = -3, b = 1/3$

C) $a \neq -3, b \neq 1/3$ D) $a = -3, b \neq 1/3$

66. The values of a,b for which the system has unique solution

A) $a \neq -3, b \in I$ B) $a = -3, b \in I$

C) $a \neq -3, b \neq 1/3$ D) $a = -3, b \neq 1/3$

67. The values of a,b for which the system has infinite solutions

A) $a = -3, b \in I$ B) $a = -3, b = 1/3$

C) $a \neq -3, b \neq 1/3$ D) $a = -3, b \neq 1/3$

Passage - VII

Consider the system of equations ax+4y+z=0, 2y+3z–1=0 and 3x–bz+2=0

68. The system of equations has a unique solution

(A) ab = 15 (B) ab $\neq$ 15

(C) ab $\in$ R (D) can not say

69. The system of equations has infinite solutions if

(A) ab = 15, a = 3, b = 5

(B) ab $\neq$ 15, a = 3, b = 5

(C) ab = 15, b = 3, a = 5

(D) ab = 10, a = 2, b = 5

70. The system of equations has no solutions if

(A) ab = 15, a $\neq$ 3, b $\neq$ 5

(B) ab = 15, a = 3, b = 5

(C) ab = 15, a $\neq$ 3

(D) ab $\neq$ 15

MATRIX MATCHING QUESTIONS

71. If $\begin{bmatrix}4a^2 & 4a & 1\\4b^2 & 4b & 1\\4c^2 & 4c & 1\end{bmatrix}\begin{bmatrix}f(-1)\\f(1)\\f(2)\end{bmatrix} = \begin{bmatrix}3a^2+3a\\3b^2+3b\\3c^2+3c\end{bmatrix}$

where f(x) is a quadratic function and $f(x) = ax^2 + bx + c$ whose maximum value occurs at a point V say (α, β). Let A be the point of intersection of $y = f(x)$ with negative x-axis, say (p,o) and point B is such that the chord AB subtends a right angle at V. Let B be (r,s). Let Δ be the area enclosed by $y = f(x)$ and the chord AB. Then

COLUMN-I	COLUMN-II
A) $\alpha + \beta =$	P) 125/3
B) $p =$	Q) -7
C) $r + s =$	R) -2
D) $\Delta =$	S) 1

72. Match the following : For 3×3 matrix, a_{ij} represents the element of i^{th} row and j^{th} column.

Column - I

A) $a_{ij}+a_{jk}+a_{ki}=0$ then

B) $a_{ij}-a_{jk}+a_{ki}=0$ then

C) $a_{ij}+a_{jk}+a_{ki}=(-1)^i-(-1)^j$

D) $(-1)^{i+j}a_{ij}+(-1)^{j+k}a_{jk}+(-1)^{k+i}a_{ki}=0$

Column - II

P) Symmetric

Q) Skew symmetric

R) All diagonal elements are zeroes

S) Value of determinant is zero

T) Trace of the matrix is zero

73. **If** $abc=1$ **and** $A=\begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix}$ **is an orthogonal matrix then**

COLUMN-I	**COLUMN-II**
A) The value of $a+b+c$ can be,	P) -1
B) The value of $ab+bc+ca$ is	Q) 0
C) The value of $a^2+b^2+c^2$ is	R) 1
D) The value of $a^3+b^3+c^3$ can be	S) 2

74. **Let** $A=\begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$

COLUMN I

(A) A^{-1}

(B) $(adjA)^{-1}$

(C) $adj(adjA)$

(D) $adj(2A)$

COLUMN II

(P) $\begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$

(Q) $2\begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$

(R) $\frac{1}{2}\begin{bmatrix} 1+\cos 2x & -\sin x \\ \sin 2x & 1+\cos 2x \end{bmatrix}$

(S) $\begin{bmatrix} 1+\cos x & \sin 2x \\ -\sin 2x & 1+\cos 2x \end{bmatrix}$

INTEGER QUESTIONS

75. **Consider three matrices** $A=\begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$, $B=\begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ **and** $C=\begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$ **then the absolute value of** $\det(A)+\det\left(\frac{ABC}{2}\right)+\det\left(\frac{A(BC)^2}{4}\right)+\det\left(\frac{A(BC)^3}{8}\right)+\text{----}\infty$ **is**

76. **If** $\lambda^3-6\lambda^2+7\lambda+2=0$ **is the characteristic equation of a matrix A then trace A + det A is**

77. $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}, AA^T=2I.$ **If** $P(\alpha,\beta)$ **divides** $Q(1,2)$ **and** $R(2,5)$ **in the ratio c : b then** $\alpha+\beta=$

78. **Let A be a matrix of order** 2×2 **such that** $A^2=O.If\ (I+A)^{100}=I+\alpha A, \alpha\in R^+$ **then numerical value of** $\alpha-96$ **is**

79. **A be the set of 3x3 matrices formed by entries 0, –1 and 1 only. There are three (1), three (–1) and three (0). The number of symmetric matrices with trace (A) = 0 is K then** $\frac{K}{6}$ **is**

80. **If** $A=\frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ **and if** $\left(A\bar{A}^T\right)=\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$ **then numerical value of** $\alpha+\beta+\gamma+\delta$ **is (here** $\bar{A}^T$ **means Transposed conjugate of A)**

81. **If A and B are any two square matrices of order 3 with** $A^3=B^3$ **and** $A^2B=B^2A\,(A\neq B)$ **then det** $\left(A^2+B^2\right)=$

82. If $A=\begin{bmatrix}2 & 1\\ -4 & -2\end{bmatrix}$ and if $I+2A+3A^2+.......\infty=\begin{bmatrix}\alpha & \beta\\ \gamma & \delta\end{bmatrix}$ then numerical value of $|\alpha+\beta+\gamma+\delta|$ is

83. If $2x+3y-3z=0$
$5x-2y+2z=19$
$x+7y-5z=5$
Then the value of $x+y-z$ is

84. If $A=\begin{pmatrix}1 & 0\\ 1 & 2\end{pmatrix}$ and $A^8=KA-\ell I$ then $K-\ell=$

85. If $A=\begin{bmatrix}1 & 0 & 0\\ i & \frac{-1+i\sqrt{3}}{2} & 0\\ 0 & 1+2i & \frac{-1-i\sqrt{3}}{2}\end{bmatrix}$ is the given matrix then trace of A^{102} will be equal to

86. If $A=\begin{bmatrix}a & b\\ 0 & a\end{bmatrix}$ is n^{th} root of $4I_2$ then the value of $a^n(\forall n\in N)$ is

87. If $A=\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}, B=\begin{bmatrix}1 & 0\\ 5 & 1\end{bmatrix}$ then the number of real values of α for which $A^2=B$ happens is/are

88. If $A=\begin{bmatrix}i & -i\\ -i & i\end{bmatrix}, B=\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$ and if $A^8=\alpha B, \alpha\in R^+$ then the numerical value of $\frac{\alpha}{32}$ is

89. If $f(x)=\begin{bmatrix}\cos x & -\sin x & 0\\ \sin x & \cos x & 0\\ 0 & 0 & 1\end{bmatrix}$ then the value of k such that $f(\theta)f(\phi)=kf(\theta+\phi)$

90. Let M be a 2×2 matrix such that $M\begin{bmatrix}1\\ -1\end{bmatrix}=\begin{bmatrix}-1\\ 2\end{bmatrix}$ and $M^2\begin{bmatrix}1\\ -1\end{bmatrix}=\begin{bmatrix}1\\ 0\end{bmatrix}$
If x_1 and $x_2\,(x_1>x_2)$ are the two values of x for which det (M - XI)=0 where I is identity matrix of order 2 then the value of $5x_1+2x_2$ is

91. The matrix $\begin{bmatrix}1 & 1 & 3\\ 5 & 2 & 6\\ -2 & -1 & -3\end{bmatrix}$ is a nilpotent matrix of index

92. If the adjoint of a 3×3 matrix P is $\begin{bmatrix}1 & 2 & 1\\ 4 & 1 & 1\\ 4 & 7 & 3\end{bmatrix}$ then sum of squares of possible values of determinant of P is

93. Let (x, y, z) be a non-zero solution of the equations $x+\lambda y+2z=0$, $2x+\lambda z=0$ and $2\lambda x-2y+3z=0$ where $\lambda\in R$ then the value of $\frac{x+y-z}{y}$ is

94. The equations $x+ky+3z=0$, $3x+ky-2z=0$, $2x+3y-4z=0$ possess a nontrivial solution then the value of 2k/33 is

SUBJECTIVE QUESTIONS

95. Show that the matrix $A=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$ satisfies $A^2=-I$. Hence, or other wise find the sixteenth power of the matrix $\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}$

96. Let A,B be two square matrices such that A+B = AB . Prove that AB =BA

97. If A & B are different matrices satisfying $A^3=B^3$ and $A^2B=B^2A$ Find $|A^2+B^2|$

98. Given, $x=cy+bz, y=az+cx, z=bx+ay$, where x,y,z are not all zero, prove that $a^2+b^2+c^2+2abc=1$ **(IIT - 1978)**

99. For what value of k does the following system of equations possess a non-trivial solution over the set of rationals $x+y-2z=0$, $2x-3y+z=0$, and $x-5y+4z=k$. Find all the solutions. (IIT - 1979)

100. Show that the system of equations $3x-y+4z=3$, $x+2y-3z=-2$ and $6x+5y+\lambda z=-3$ has at least one solution for any real number $\lambda\neq-5$. Find the set of solutions, if $\lambda=-5$. (IIT - 1983)

101. Let λ and α be real. Find the set of all values of λ for which the system of linear equations $\lambda x+(\sin\alpha)y+(\cos\alpha)z=0$, $x+(\cos\alpha)y+(\sin\alpha)z=0$, and $-x+(\sin\alpha)y-(\cos\alpha)z=0$ has a non - trivial solution. For $\lambda=1$, find all values of α (IIT 1993)

102. If M is a 3×3 matrix, where $M^TM=I$ and det $(M)=1$, then prove that det $(M-I)=0$. (IIT 2004)

103. If matrix $A=\begin{bmatrix} a & b & c\\ b & c & a\\ c & a & b\end{bmatrix}$, where a, b, c are real positive numbers, abc = 1 and $A^TA=I$, then find the value of $a^3+b^3+c^3$. [IIT-2003]

KEY
LEVEL -V

SINGLE ANSWER QUESTIONS

1. D	2.C	3.D	4.B
5.D	6.D	7.D	8.A
9.A	10.D	11.A	12.C
13.A	14.B	15.B	16.B
17.A	18.D	19.B	20.C
21.C	22.A	23.D	24.A
25.A	26.C	27.C	28.A
29.D	30.A	31.A	

MULTI ANSWER QUESTIONS

32.AB	33.ABC	34.ABC	35.AB
36.ACD	37.ABCD	38.ABC	39.AB
40.ABC	41.BC	42.ABC	43.AB
44.BC	45.AD	46.ABC	47.BCD
48.ABCD	49.AC		

COMPREHENSION QUESTIONS

50. C	51. C	52. B	53. B
54. D	55. D	56. A	57. A
58. C	59. A	60. B	61. B
62. B	63. C	64. A	65. D
66. A	67. B	68B	69.A
70.C			

MATRIX MATCHING QUESTIONS

71. A-S; B-R; C-Q; D-P
72. A-Q,R,S,T; B-P,R,T; C-R,T;D-Q,R,S,T
73. A-P,R; B-Q; C-R, D-S
74. A-R; B-S; C-P, D-Q

INTEGER QUESTIONS

75. 2	76. 4	77. 5	78. 4
79. 6	80. 2	81. 0	82. 2
83. 1	84. 1	85. 3	86. 4
87. 0	88. 1	89. 1	90. 8
91. 3	92. 8	93. 5	94. 1

SUBJECTIVE QUESTIONS

95. 2^8I 97. Not exist

99. $m<\dfrac{-15}{2}$ or $m>30$

100. $K=0$

102. $\alpha=n\pi$ or $n\pi+\dfrac{\pi}{4}$

HINTS
LEVEL -V

SINGLE ANSWER QUESTIONS

1. Clearly option (D) Satisfies the given conditions.

2. $A^2=\begin{bmatrix}1 & 0\\ -1 & 7\end{bmatrix}\begin{bmatrix}1 & 0\\ -1 & 7\end{bmatrix}=\begin{bmatrix}1 & 0\\ -8 & 49\end{bmatrix}$

$8A+\alpha I=\begin{bmatrix}8 & 0\\ -8 & 56\end{bmatrix}+\begin{bmatrix}\alpha & 0\\ 0 & \alpha\end{bmatrix}$

$=\begin{bmatrix}8+\alpha & 0\\ -8 & 56+\alpha\end{bmatrix}$

$8+\alpha=1$

$+\alpha=-7$

3. $AB=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}_{3\times3}=0$

4. $AB=\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}=\begin{pmatrix}a & 2b\\ 3a & 4b\end{pmatrix}$

$$BA=\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$=\begin{pmatrix} a & 2b \\ 3a & 4b \end{pmatrix} AB=BA \Rightarrow a=b$$

5. $A^2=4A-5I$

$A^3=11A-20I$

$A^6=A^3.A^3=44A-205I$

6. Given, $A=\begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B=\begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$ and $A^2=B$

$\Rightarrow \quad A^2=\begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}=\begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix}$

$\therefore \quad A^2=B$

$\Rightarrow \quad \begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$

or $\quad \alpha^2=1$ and $\alpha+1=5$

which is not possible at the same time

$\therefore \quad$ no real values of α.

7. $(A+B)^r=A^r+rc_1A^{r-1}B+.......+B^r=0$ if 'r' is L.C.M of m,n

8. The matrices A^3 and A^2 can be computed as usual.

$\therefore A^3-2A^2+3A-I$

$$=\begin{bmatrix} 24 & 14 & 34 \\ 18 & 8 & 26 \\ 4 & 0 & 6 \end{bmatrix}-2\begin{bmatrix} 9 & 1 & 13 \\ 5 & 3 & 7 \\ 0 & 2 & 0 \end{bmatrix}$$

$$+3\begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & 3 \\ 1 & -1 & 1 \end{bmatrix}-\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore B'=\begin{bmatrix} 8 & 14 & 7 \\ 21 & 1 & -7 \\ 14 & 21 & 8 \end{bmatrix}$$

9. Let $X=\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$

$\therefore X^1.A.X=0$

$\Rightarrow a_{11}X_1^2+a_{22}X_2^2+a_{33}X_3^2+(a_{12}+a_{21})X_1X_2$

$+(a_{13}+a_{31})X_1X_3+(a_{23}+a_{32})X_2X_3=0$

This is true X_i

$a_{11}=a_{22}=a_{33}=0$

$a_{12}+a_{21}=0$

$a_{13}+a_{31}=0$

$a_{23}+a_{32}=0$

$\Rightarrow a_{32}=-a_{23}=-(-2009)=2009$

10. $A^TA=I \Rightarrow C^n=(ABA^T)(ABA^T)$ - - - (ABA^T) (n times)

$=AB^nA^T$

$\therefore A^TC^nA=A^TAB^nA^TA=B^n=\begin{bmatrix} 1 & 0 \\ -n & 1 \end{bmatrix}$

11. Given $a,b,c \in \{\omega,\omega^2\}$

Let $M=\begin{vmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{vmatrix}$ but

$M=1-(a+c)\omega+ac\omega^2$ observe the table

a	c	$a+c$	ac
ω	ω^2	-1	1
ω	ω	2ω	ω^2
ω^2	ω	-1	1
ω^2	ω^2	$2\omega^2$	ω

The possible value of $1-(a+c)\omega+ac\omega^2$ are

$1+\omega+\omega^2=0, \quad 1-2\omega^2+\omega=-3\omega^2,$

$1+\omega+\omega^2=0, \quad 1-2\omega^3+\omega^3=0$

Thus the det $A\neq 0$ if and only if $a=c=\omega$

Thus b can take two values $b=\omega,\omega^2$.

Hence the number of matrices is 2.

12. $|A|=\alpha\beta\gamma-(8\alpha+4\beta+2\gamma)+28$

$=60-20+28$

$(\therefore \alpha\beta\gamma=product\ of\ roots=60)$

=68 , $8\alpha+4\beta+2\gamma=20$ as p lines on the plane

13. adj $A=\begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ ac-b & -c & 1 \end{bmatrix}$

$|A|=1$

$$\Rightarrow A^{-1}=\begin{bmatrix}1 & 0 & 0\\ -a & 1 & 0\\ ac-b & -c & 1\end{bmatrix}$$

14. we must have $AA^{-1}=I$
(3,1)th entry of $AA^{-1}=0$ =(1, 3)th entry of AA^{-1}

$\Rightarrow 3\times\frac{1}{2}+a\times(-4)+1\times\frac{5}{2}=0$

$=0\times\frac{1}{2}+1+c+2\times\frac{1}{2}=0$

$\Rightarrow -4a+4=0$ and c+1 =0

$\Rightarrow$ a =1 and c = - 1.

15. Adj $\left(Q^{-1}.B.P^{-1}\right)=Adj\left(Q^{-1}.B\right)P^{-1}$

$=AdjP^{-1}.Adj\left(Q^{-1}.B\right)$

$=AdjP^{-1}.Adj.B.AdjQ^{-1}$

$=PAQ$.

16. $(I-A)f(A)=I+A$

$\Rightarrow f(A)=(I-A)^{-1}(I+A)$

$\left(I+A+A^2\right)(I+A)$

$=I+2A+2A^2\left(\therefore A^3=0\right)$

17. $I+P+P^2+-+P^n=0$

$\Rightarrow P^{-1}+I+P+--+P^{n-1}=0$

$\Rightarrow p^{-1}=-\left(I+P+-+P^{n-1}\right)=-\left(-P^n\right)=P^n$

18. $\{F(x).G(y)\}^{-1}=G^{-1}(y).F^{-1}(x) if$

$|f(x)|\neq 0.|G(x)|\neq 0$

Here , $|f(x)|=1.|G(y)|=1$

Also , $F^{-1}(x)=\begin{bmatrix}\cos x & \sin x & 0\\ -\sin x & \cos x & 0\\ 0 & 0 & 1\end{bmatrix}=F(-x)$

and $G^{-1}(y)=\begin{bmatrix}\cos y & 0 & -\sin y\\ 0 & 1 & 0\\ \sin y & 0 & \cos y\end{bmatrix}=G(-y)$

$\Rightarrow\left[F(x).G(y)\right]^{-1}=G^{-1}(y)F^{-1}(x)$

$=G(-y)F(-x)$

19. We have

$A^n=\begin{pmatrix}\cos n\alpha & \sin n\alpha\\ -\sin n\alpha & \cos n\alpha\end{pmatrix}$

We have

$\cos\alpha+\cos2\alpha+\cos3\alpha+\cos4\alpha=0$

$\therefore 5\alpha=\pi$

And $\sin\alpha+\sin2\alpha+\sin3\alpha+\sin4\alpha$

$=4\cos\frac{\pi}{5}\cos\frac{\pi}{10}=a(\text{say})$

$|B|=a^2=16\cos^2\frac{\pi}{5}\cos^2\frac{\pi}{10}>0$

$\therefore$ B is non singular

20. Every row vecor is a unit vector and every pair of vectors are orthoganal. It is orthogonal.

21. $\begin{bmatrix}a & i\\ i & b\end{bmatrix}.\begin{bmatrix}a & i\\ i & b\end{bmatrix}=\begin{bmatrix}a^2-1 & (a+b)i\\ (a+b)i & b^2-1\end{bmatrix}$

$\therefore a^2-1=a,\ a+b=1$

$a=\frac{1\pm\sqrt{5}}{2}.b=\frac{1\mp\sqrt{5}}{2}$

22. We have $PP^T=I$, $A^5=A$

$Q^{2021}=PA^{2021}P^T$

$\therefore X=A^{2021}=A^2$

23. $\Delta=\begin{vmatrix}1 & 2\omega & 3\omega^2\\ 2 & 3\omega & \omega^2\\ 3 & \omega & 2\omega^2\end{vmatrix}=\begin{vmatrix}1 & 2\omega & 3\omega^2\\ 0 & -\omega & -5\omega^2\\ 0 & -5\omega & -7\omega^2\end{vmatrix}$

= 1(7–25) = –18

$\Delta_1=\begin{vmatrix}1-\omega^2 & 2\omega & 3\omega^2\\ \omega^2-\omega & 3\omega & \omega^2\\ \omega-1 & \omega & \omega^2\end{vmatrix}=$

$(\omega-1)\begin{vmatrix}-(1+\omega) & 2 & 3\\ \omega & 3 & 1\\ 1 & 1 & 2\end{vmatrix}=(\omega-1)\begin{vmatrix}0 & 6 & 6\\ \omega & 3 & 1\\ 1 & 1 & 2\end{vmatrix}$

$=(\omega-1)(-6(2\omega-1)+6(\omega-3))$

$=(\omega-1)(-6\omega-12)$

$=6(\omega-1)(\omega+2)=-6\left(\omega^2+\omega-2\right)$=18

24. The system can have unique or infinitely many solutions. If can not have two distinct solutions.

25. $Here, \Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 5 & 5 & 9 \end{vmatrix} \neq 0$

26. $\begin{vmatrix} 6 & 5 & \lambda \\ 3 & -1 & 4 \\ 1 & 2 & -3 \end{vmatrix} = 0$

27. $\begin{vmatrix} K & 4 & 1 \\ 4 & K & 2 \\ 2 & 2 & 1 \end{vmatrix} = 0$

28. Clearly $A^2 = A^4 = A^6 ...... = I$

$x = 2,4,8..... \therefore \sum\left(\cos^x\theta + \sin^x\theta\right)$

$= \left(\cos^2\theta + \cos^4\theta + ...\right) + \left(\sin^2\theta + 8^4\theta + ...\right)$

$= \cot^2\theta + \sec^2\theta \geq 2$

29. We have $P^T = 2P + I \Rightarrow P^T - 2P = I - (1)$

$\Rightarrow P - 2P^T = I - (2)$

$\therefore$ From (1) *and* (2)

$P = -I \Rightarrow (P+I)X = 0 \Rightarrow PX = -X$

30. $\begin{vmatrix} 2a & -2 & 3 \\ 1 & a & 2 \\ 2 & 0 & a \end{vmatrix} = 0$

$2a^3 - 4a - 8 = 0$

$a^3 - 2a - 4 = 0$

$(a - 2)(a^2 + 2a + 2) = 0$

$a = 2$

31.

$2x + 3y + 1 = 0$

$2x + y - 1 = 0$ on solving $x = 1, y = -1$

If equations are consistent, then

$a(1) + 2(-1) - b = 0$

$a - b - 2 = 0$

MULTI ANSWER QUESTIONS

32. $A^2 = -kI$

$\begin{bmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{bmatrix}\begin{bmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{bmatrix} = \begin{bmatrix} -K & 0 \\ 0 & -K \end{bmatrix}$

33. $A^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$

$\Rightarrow A^3 = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 6 & 1 \end{bmatrix}$

$A^{2012} = \begin{bmatrix} 1 & 0 \\ 4024 & 1 \end{bmatrix}$

$\Rightarrow a = d, a + b + c + d = 4026$

34. $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

35. $\begin{vmatrix} 5-\lambda & 7 & 3 \\ 1 & 5-\lambda & 2 \\ 3 & 2 & 1-\lambda \end{vmatrix} = 0$

$\Rightarrow (5-\lambda)(5 - 6\lambda + \lambda^2 - 4)$

$-7(-5-\lambda) + 3(2 - 15 + 3\lambda) = 0$

$\Rightarrow 5\lambda^2 - 30\lambda + 5 - \lambda^3 + 6\lambda^2$

$-\lambda + 35 + 7\lambda - 39 + 9\lambda = 0$

$\Rightarrow -\lambda^3 + 11\lambda^2 - 15\lambda + 1 = 0$

$\Rightarrow \lambda^3 - 11\lambda^2 + 15\lambda + 1 = 0$

$\therefore A^3 - 11A^2 + 15A - I = 0 \Rightarrow K = 11, l = 15$

36. Write the characteristic equation

37. $AB = \begin{bmatrix} \cos(\alpha + 2\beta) & \sin(\alpha + 2\beta) \\ \sin(\alpha + 2\beta) & -\cos(\alpha + 2\beta) \end{bmatrix}$

$\Rightarrow (AB)^2 = I$

$\Rightarrow (AB)(AB) = I \Rightarrow A^{-1} = BAB$

$(AB)^{-1} = AB$

$\Rightarrow B^{-1}A^{-1} = AB$

$\Rightarrow BB^{-1}A^{-1} = BAB \Rightarrow A^{-1} = BAB$

$BA^4B = A^{-1} \Rightarrow B^{-1}BA^4B = B^{-1}A^{-1}$

$\Rightarrow A^4B = (AB)^{-1} = AB$

$\Rightarrow A^4 = A \Rightarrow \cos 4\alpha = \cos\alpha, \sin 4\alpha = \sin\alpha$

$4\alpha = 2\pi + \alpha \Rightarrow \alpha = \dfrac{2\pi}{3}$

38. $Tr(A) = -1,\ Tr(A) = -1$

$\Rightarrow A^{2012} = (I)^{1006} = I \Rightarrow$ Tracef of A^{2012} is 3.

A is involutary.

39. Every row vector is unit vector and the vectors of any two rows are orthogonal.

$SAS^{-1} = \begin{bmatrix} 2a & 0 & 0 \\ 0 & 2b & 0 \\ 0 & 0 & 2c \end{bmatrix} \therefore$ diagonal also

Invertible

40. $\left(A^{-1}\right)^{-1}=A$ and $(AB)^{-1}=B^{-1}A^{-1}$

41. $A=\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}$

$|A|=-1$

$\therefore$ Adj.A $=\begin{bmatrix}0&0&-1\\0&-1&0\\-1&0&0\end{bmatrix}$

Hence $A^{-1}=A$

42. $B=P^{-1}AP,\ B^2=(P^{-1}AP)(P^{-1}AP)$

$=P^{-1}A^2P \Rightarrow B^n=P^{-1}A^nP\ \forall n\in N$

$|B|=|P^{-1}AP|=|P^{-1}||A||P|=|A|$

$\therefore if\ |A|\neq 0$ then $|B|\neq 0$

$P^{-1}(A-\lambda I)P=P^{-1}AP-P^{-1}\lambda P\ =B-\lambda I$

43. $(A+B)^{-1}(A-B)(A-B)^T\left((A+B)^T\right)^{-1}$

$=(A+B)^{-1}(A-B)(A+B)(A-B)^{-1}$

44. $\left(-A^T\right)^{-1}=\dfrac{adj(-A)}{|-A|}=$

$\dfrac{(-1)^{n-1}adj(A)}{(-1)^n|A|}=\dfrac{adj(A)}{-|A|}$

$=A^{-1}(\forall n)$

given $A^n=0$ now

$(I-A)(I+A+A^2......+A^{n-1})=I-A^n=I$

$\therefore (I-A)^{-1}=I+A+.....+A^{n-1}$

45. $\det(adJ\ P)=(\det P)^2 \Rightarrow \det P=2\ or -2$

46. $(A+B)C=(A+B)(A+B)^{-1}(A-B)$

$=A-B$

$C^T=(A-B)^T\left((A+B)^{-1}\right)^T$

$=(A-B)^T\left((A+B)^T\right)^{-1}$

$=(A-B)^T(A-B)^{-1}=(A+B)(A-B)^{-1}$

$C^T(A+B)C$

$=(A+B)(A-B)^{-1}(A-B)=A+B$ -------1

Taking transpose

$\Rightarrow C^T(A+B)^T\left(C^T\right)^T=(A+B)^T$

$\Rightarrow C^T(A-B)C=A-B$ ----------2

$1+2\Rightarrow C^T.2A.C=2A\Rightarrow C^TAC=A$

47. $A^2=I\Rightarrow A^{-1}=A\Rightarrow adj\ A=A$

48. $((I+S)(I-S)^{-1})^T(I+S)(I-S)^{-1}$

$=((I-S)^{-1})^T(I+S)^T(I+S)(I-S)^{-1}$

$=(I+S)^{-1}(I-S)(I+S)(I-S)^{-1}$

$=(I+S)^{-1}(I+S)(I-S)(I-S)^{-1}=I$

$\therefore (I+S)(I-S)^{-1}$ is orthogonal.

49 $\Delta=\begin{vmatrix}a&b&c\\b&c&a\\c&a&b\end{vmatrix}=0$

$\Rightarrow a+b+c=0$ or $a=b=c$

$a>b>c \Rightarrow a+b+c=0$

$\therefore$ 1 is a root of $at^2+bt+c=0$

Other root is $\dfrac{c}{a}>0$

COMPREHENSION QUESTIONS

50,51,52. $B_0^{\,2}=I$ and $A_0^{\,2}=A_0$

$\therefore$ det $(A_0+A_0^{\,2}+.........10terms)$

$=\det(10A_0)=1000$

$X\ 0=0$

$B_1=B_2=..........B_{49}=B_0$

$A_0X=C_1$ has no solution.

53,54,55. $A^{50}=A^{48}+A^2-I$

$=25A^2-24I=\begin{bmatrix}1&0&0\\25&1&0\\25&0&1\end{bmatrix}\Rightarrow \det A^{50}=1$

and $Trace\ A^{50}=3$

56,57,58 $A^2=\begin{bmatrix}2&3\\1&2\end{bmatrix}\begin{bmatrix}2&3\\1&2\end{bmatrix}$

$=\begin{bmatrix}4+3&6+6\\2+2&3+4\end{bmatrix}=\begin{bmatrix}7&12\\4&7\end{bmatrix}$

$=\begin{bmatrix}7&12\\4&7\end{bmatrix}-\begin{bmatrix}8&12\\4&8\end{bmatrix}+\begin{bmatrix}1&0\\0&1\end{bmatrix}=O$

$\therefore A^2 - 4A + I = 0$

$A^{-1}(A^2) - 4A^{-1}A + A^{-1}I = 0$

$A = 4I - A^{-1} \Rightarrow A^{-1} = 4I - A$

$A^{-1} = 4\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} - \begin{bmatrix}2 & 3\\1 & 2\end{bmatrix} = \begin{bmatrix}4 & 0\\0 & 4\end{bmatrix} - \begin{bmatrix}2 & 3\\1 & 2\end{bmatrix}$

$= \begin{bmatrix}2 & -3\\-1 & 2\end{bmatrix}$

Clearly a = -4, b = 1

$\therefore \int_{-4}^{4} x^3 \cos x dx = 0$

Also $\dfrac{a+4b}{4a-b} = \dfrac{-4+4}{4\times -4-1} = 0$

59,60,61

$A = \begin{bmatrix}2 & -1 & 3\\1 & 1 & -2\\1 & 1 & 1\end{bmatrix}$

$\Rightarrow |A| = 2(1+2) + 1(1+2) + 3(1-1) \neq 0$

The solution is unique

. If the system does not have a unique solution the value of the determinant of coefficients =0

$\therefore \begin{vmatrix}2 & -1 & 1\\1 & 3 & 2\\3 & 2 & k\end{vmatrix} = 0 \Rightarrow k = 3$

. The required conditions are $|A| = 0$ and (Adj A) B = 0

$\therefore \begin{vmatrix}1 & 1 & 1\\1 & 2 & 3\\1 & 2 & \lambda\end{vmatrix} = 0,$ and

$\begin{bmatrix}2\lambda-6 & 2-\lambda & 1\\-\lambda+3 & \lambda-1 & -2\\0 & -1 & 1\end{bmatrix}\begin{pmatrix}6\\10\\\mu\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$

i.e., $(2\lambda-6)+(3-\lambda)+0=0$ and 0.6 - 10 + $\mu = 0$

$\Rightarrow \lambda = 3, \mu = 10$

62,63,64

$\begin{bmatrix}1 & 0 & 0\\2 & 1 & 0\\3 & 2 & 1\end{bmatrix}\begin{bmatrix}a\\b\\c\end{bmatrix} = \begin{bmatrix}1\\0\\0\end{bmatrix}$

$\Rightarrow a = 1, 2a+b = 0 \Rightarrow b = -2$

$3a+2b+c = 0 \Rightarrow c = 1$

$\begin{bmatrix}1 & 0 & 0\\2 & 1 & 0\\3 & 2 & 1\end{bmatrix}\begin{bmatrix}d\\e\\f\end{bmatrix} = \begin{bmatrix}2\\3\\0\end{bmatrix}$

$\Rightarrow d = 2, 2d+e = 3 \Rightarrow e = -1$

$3d+2e+f = 0 \Rightarrow 6-2+f = 0 \Rightarrow f = -4$

$\begin{bmatrix}1 & 0 & 0\\2 & 1 & 0\\3 & 2 & 1\end{bmatrix}\begin{bmatrix}p\\q\\r\end{bmatrix} = \begin{bmatrix}2\\3\\1\end{bmatrix}$

$\Rightarrow p = 2, 2p+q = 3 \Rightarrow q = -1$

$3p+2q+r = 1 \Rightarrow r = -3$

$B = \begin{bmatrix}1 & 2 & 2\\-2 & -1 & -1\\1 & -4 & -3\end{bmatrix}$

$|B| = 1(-1) - 2(7) + 2(9) = 3$

$C = \begin{bmatrix}1 & 0 & 0\\2 & 1 & 0\\3 & 2 & 1\end{bmatrix}\begin{bmatrix}1 & 2 & 2\\-2 & -1 & -1\\1 & -4 & -3\end{bmatrix}$

$= \frac{1}{3}\begin{bmatrix}1 & 2 & 2\\0 & 3 & 3\\0 & 0 & 1\end{bmatrix}$ $|A| = 1, |C| = 1$

$\begin{matrix}+(-1) & -7 & +9\\-2 & +(-5) & -(-6)\\+0 & -3 & +3\end{matrix}$

$B^{-1} = \frac{1}{3}\begin{bmatrix}-1 & -2 & 0\\-7 & -5 & -3\\9 & 6 & 3\end{bmatrix}$

Sum of elements = 0

65,66,67

$\Delta = -14a - 42, \Delta = -8ab - 9b + 6a + 13$

$= -14(a+3) = 15b - 5at\ a = -3$

$\therefore a \neq -3, b \in R$, unique solution

$a=-3, b\neq \frac{1}{3}$, no solution

$a=-3, b=\frac{1}{3}$, infinite solutions.

68,69,70

The system can be written as
ax + 4y + z = 0,   0x + 2y + 3z = 1
& 3x + 0y – bz = -2,

$$\Rightarrow \ D=\begin{vmatrix} a & 4 & 1 \\ 0 & 2 & 3 \\ 3 & 0 & -b \end{vmatrix}=30-2ab$$

$$D_x=\begin{vmatrix} 0 & 4 & 1 \\ 1 & 2 & 3 \\ -2 & 0 & -b \end{vmatrix}=4b-20$$

$$D_y=\begin{vmatrix} a & 0 & 1 \\ 0 & 1 & 3 \\ 3 & -2 & -b \end{vmatrix}=-ab+6a-3$$

$$D_z=\begin{vmatrix} a & 4 & 0 \\ 0 & 2 & 1 \\ 3 & 0 & -2 \end{vmatrix}=-4a+12$$

Now the system will have unique solution if $D \neq 0 \Rightarrow ab \neq 15$

If ab = 15, let us write D_x, D_y, D_z, we get

$D_x = 4b-20$, $D_y=6a-18$, $D_z=-4a+12$

we note that there is a value of a for which D_x, D_y, D_z vanish simultaneously which is obviously a = 3; Also for $a \neq 3$ all the determinants become non-zero. Thus the system will have infinite solutions if ab = 15, a=3 and will have no solutions if ab = 15, $a \neq 3$.

MATRIX MATCHING QUESTIONS

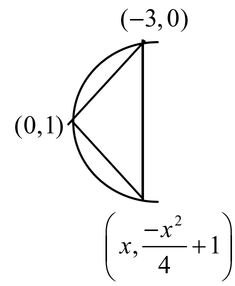
71.
$$4a^2 f(-1)+4af(1)+f(2)=3a^2+3a$$
$$4b^2 f(-1)+4bf(1)+f(2)=3b^2+3b$$

$$4f(-1)(a^2-b^2)+4f(1)(a-b)$$
$$=3(a^2-b^2)+3(a-b)$$
$$4f(-1)(a+b)+4f(1)=3(a+b)+3$$
$$4f(-1)=3 \qquad 4f(1)=3$$
$$f(-1)=\frac{3}{4}, \qquad f(1)=\frac{3}{4},$$

$$f(2)=0$$
$$f(2)=0$$
$$4a+2b+c=0$$
$$a-b+c=\frac{3}{4}$$
$$a+b+c=\frac{3}{4}$$

$$2b=0 \Rightarrow b=0$$
$$4a+c=0$$
$$a+c=\frac{3}{4}$$

$$3a=-\frac{3}{4}\Rightarrow a=-\frac{1}{4}$$
$$c=\frac{3}{4}+\frac{1}{4}=1$$
$$f(x)=-\frac{1}{4}x^2+1$$
$$(\alpha,\beta)=(0,1)$$
$$A=(-2,0)\Rightarrow P=-2$$
$$\frac{-\frac{x^2}{4}}{x}\times\frac{1}{2}=-1\Rightarrow x=8 \qquad B(8,-15)$$



$$y-0=-\frac{3}{2}(x+2)$$
$$\Rightarrow 2y=-3x-6\Rightarrow y=-\frac{3}{2}x-3$$
$$\Rightarrow 3x+2y+6=0$$
$$y_1-y_2=-\frac{x^2}{4}+1+\frac{3}{2}x+3=-\frac{x^2}{4}+\frac{3x}{2}+4$$

$$\Delta=\frac{\left(\frac{9}{4}-4\times-\frac{1}{4}\times4\right)^{3/2}}{6\times\frac{1}{16}}=\frac{125}{8}\times\frac{8}{3}$$

72. Put $i=j=k$ and then $j=k$, $k=i$ etc.

73. $a^2+b^2+c^2=1,\ ab+bc+ca=0$

$\Rightarrow (a+b+c)^2=1\Rightarrow a+b+c=\pm1$

$a^3+b^3+c^3-3abc$

$=(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$

$=\pm1$

$\Rightarrow a^3+b^3+c^3=\pm1+3=4\,or\,2$

74. $A=\begin{bmatrix}1 & \tan x\\ -\tan x & 1\end{bmatrix}$

$\therefore adj(A)=\begin{bmatrix}1 & -\tan x\\ \tan x & 1\end{bmatrix}$

$\therefore A^{-1}=\dfrac{adj(A)}{|A|}$

$=\dfrac{1}{1+\tan^2 x}\begin{bmatrix}1 & -\tan x\\ \tan x & 1\end{bmatrix}$

$=\begin{bmatrix}\cos^2 x & -\sin x\cos x\\ \sin x\cos x & \cos^2 x\end{bmatrix}$

$=\dfrac{1}{2}\begin{bmatrix}1+\cos 2x & -\sin 2x\\ \sin 2x & 1+\cos 2x\end{bmatrix}$

$\therefore (A)\to(R)$

$Adj(AdjA)=\begin{bmatrix}1 & -\tan x\\ \tan x & 1\end{bmatrix}=A$

$\therefore (AdjA)^{-1}=\dfrac{Adj(AdjA)}{|AdjA|}$

$=\dfrac{1}{2}\begin{bmatrix}1+\cos 2x & \sin 2x\\ -\sin 2x & 1+\cos 2x\end{bmatrix}$

$\therefore (B)\to(S); and (C)\to(P)$

$2A=2\begin{bmatrix}1 & -\tan x\\ \tan x & 1\end{bmatrix}$

$\therefore (adj\,2A)=2^{2-1}\,adj\,(A)$

$=2\begin{bmatrix}1 & -\tan x\\ \tan x & 1\end{bmatrix}\therefore D-Q$

INTEGER QUESTIONS

75. $BC=I$

$\Rightarrow \det(A)+\det\left(\frac{A}{2}\right)+\det\left(\frac{A}{4}\right)+\text{-------}\infty$

$\Rightarrow \det(A)\left(1+\frac{1}{2}+\frac{1}{4}+---\infty\right)=2$

76. In the characteristic equation sum of the roots gives trace and product of roots gives det of the matrix

77. $\left(\dfrac{2c+b}{b+c}\quad \dfrac{5c+2b}{b+c}\right)=(\alpha,\beta)$

$\alpha+\beta=\dfrac{7c+3b}{b+c}$

$\begin{bmatrix}a & b\\ c & d\end{bmatrix}\begin{bmatrix}a & c\\ b & d\end{bmatrix}=\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}$

$a^2+b^2=2$

$c^2+d^2=2$

$ac+bd=0$

$\Rightarrow b=c$

78. $\alpha=100$

$\therefore \alpha-96=4$

79. The elements 0, –1, 1 can be arranged in the principal diagnol in 3. ways (and remaining places can be filled 3! ways corresponding to each case $(\because a_{ij}=a_{ji})$

:. No of matrices $=3!\,3!=36\Rightarrow \dfrac{K}{6}=6$

80. $AA^{-T}=\dfrac{1}{\sqrt3}\begin{bmatrix}1 & 1+i\\ 1-i & -1\end{bmatrix}\dfrac{1}{\sqrt3}\begin{bmatrix}1 & 1+i\\ 1-i & -1\end{bmatrix}$

$=\dfrac{1}{3}\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

$\alpha+\beta+\gamma+\delta=2$

81. $(A^2+B^2)(A-B)$

$= A^3 - A^2B + B^2A - B^3 = 0$

If $\left|A^2 + B^2\right| \neq 0$ then $A - B = 0 \Rightarrow A = B$

$\therefore \left|A^2 + B^2\right| = 0$

82. Let $S = I + 2A + 3A^2 + ...\infty$

$AS = \quad A + 2A^2 + ...\infty$

$(I - A)S = I + A + A^2 + ...\infty$

$= \dfrac{I}{I - A} = I(I - A)^{-1}$

$I - A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ -4 & -2 \end{bmatrix}$

$= \begin{bmatrix} -1 & -1 \\ 5 & 3 \end{bmatrix}$

$S = \dfrac{(I - A)^{-1}}{I - A}$

$(I - A)^{-1} = \dfrac{1}{2}\begin{bmatrix} 3 & 1 \\ -5 & -1 \end{bmatrix}$

$= \left((I - A)^{-1}\right)^2$

$= \dfrac{1}{4}\begin{bmatrix} 3 & 1 \\ -5 & -1 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ -5 & -1 \end{bmatrix}$

$= \dfrac{1}{4}\begin{bmatrix} 4 & 2 \\ -10 & -4 \end{bmatrix} = \begin{bmatrix} 1 & \dfrac{1}{2} \\ -\dfrac{5}{2} & -1 \end{bmatrix}$

$|\alpha + \beta + \gamma + \delta| = 2$

83. $2x + 3y - 3z = 0$ ...(1)

$5x - 2y + 2z = 19$ ...(2)

$x + 7y - 5z = 5$ ...(3)

$5 \times (3) - (2) \Rightarrow 37y - 27z = 6$ ...(4)

$2 \times (3) - (1) \Rightarrow 11y - 7z = 10$ ...(5)

Solving (4) & (5) we have

$y = 6, z = 8$

from (1) $x = 3$

$x + y - z = 3 + 6 - 8 = 1$

84. $\begin{vmatrix} 1-\lambda & 0 \\ 1 & 2-\lambda \end{vmatrix} = 0 \quad \Rightarrow (1-\lambda)(2-\lambda) = 0$

$\Rightarrow 2 - 3\lambda + \lambda^2 = 0 \quad \Rightarrow A^2 = 3A - 2I$

$A^4 = (3A - 2I)(3A - 2I)$

$= 9A^2 - 12A + 4I \quad = 15A - 14I$

$A^8 = (15A - 14I)(15A - 14I)$

$= 225(3A - 2I) - 420A + 196I$

$= 255A - 254I \quad \Rightarrow K - l = 1$

85. $A^3 = I \qquad \Rightarrow A^{102} = I \Rightarrow trace = 3$

86. $\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} a^n & b^n \\ 0 & a^n \end{bmatrix} \Rightarrow a^n = 4$.

87. Given, $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$ and $A^2 = B$

$\Rightarrow \quad A^2 = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix}$

$\therefore \quad A^2 = B$

$\Rightarrow \quad \begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$

or $\qquad \alpha^2 = 1$ and $\alpha + 1 = 5$

which is not possible at the same time

$\therefore$ no real values of α.

88. $A^2 = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}\begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$

$= \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = 2\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

$A^4 = 4\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

$= 4\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = 4\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

$A^8 = 16\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

$= 16\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = 32\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 32B$

89. Clearly $f(\theta).f(\phi) = f(\theta + \phi) \Rightarrow k = 1$

90. Let $m = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$\therefore a - b = -1, c - d = 2$

$-a + 2b = 1, -c + 2d = 0 \qquad \Rightarrow m = \begin{bmatrix} -1 & 0 \\ 4 & 2 \end{bmatrix}$

$\therefore |m - xI| = 0$

$\Rightarrow (x - 2)(x + 1) = 0 \Rightarrow x_1 = +2 x_2 = -1$

$5x_1 + 2x_2$

91. $\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$A^3 = 0 \quad \therefore\ Index = 3$

92. $|P|^2 = \begin{vmatrix} 1 & 2 & 1 \\ 4 & 1 & 1 \\ 4 & 7 & 3 \end{vmatrix} = 4 \Rightarrow |P| = \pm 2$

93. $\begin{vmatrix} 1 & \lambda & 2 \\ 2 & 0 & \lambda \\ 2x & -2 & 3 \end{vmatrix} = 0 \Rightarrow$

$1\ (1(2\lambda) - x(6 - 2\lambda^2) + 2(-4) = 0$

$= 2\lambda^3 - 4\lambda - 8 = 0 \quad = \lambda^3 - 2\lambda - 4 = 0$

Let z = K $\Rightarrow\ x = \dfrac{-\lambda K}{2}$,

2y = $2\lambda x - \dfrac{\lambda K}{2} + 3K$

$\Rightarrow (\lambda - 2)(x^2 + 2^2\lambda + 2) = 0$

$= \lambda = 2$

3K $-x^2$K = $y = \dfrac{3K - \lambda^2 K}{2}$

$\dfrac{\dfrac{-\lambda K}{2} + \dfrac{3K - \lambda^2 K}{2} - K}{\dfrac{3K - \lambda^2 K}{2}}$

$= \dfrac{-\lambda + 3 - \lambda^2 - 2}{3 - \lambda^2} = \dfrac{-\lambda^2 - \lambda + 1}{3 - \lambda^2} = \dfrac{-5}{-1} = 5$

94. Coefficient of det = $0 \Rightarrow \begin{vmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{vmatrix} = 0$

$\Rightarrow (-4k + 6) - k(-12 + 4) + 3(9 - 2k) = 0$

$\Rightarrow -4k + 6 + 12k - 4k + 27 - 6k = 0$

$\Rightarrow -2k + 33 = 0$

$2K = 33$

$\dfrac{2K}{33} = 1$

SUBJECTIVE QUESTIONS

95. Let $B = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \Rightarrow B = A + I$

$\Rightarrow B^2 = 2A \Rightarrow B^{16} = 2^8 I$

96. Since $AB - A - B = 0_{n,}$ by adding I_n to both sides and factoring we obtain $(I_n - A)(I_n - B) = I_n$.

follows that $I_n - A$ is invertible, and its inverse is $I_n - B$. Hence $(I_n - B)(I_n - A) = I_n$, which implies $BA - A - B = 0_{n,}$. Consequently, $BA = A + B = AB$.

97. We have

$(A^2 + B^2)(A - B) = A^3 - A^2B + AB^2 - B^3$.

since $A \neq B$, this shows that $A^2 + B^2$ has a zero divisor. Hence it is not invertible, so its determinant is 0.

98 $\therefore \begin{vmatrix} -1 & c & b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0 \Rightarrow a^2 + b^2 + c^2 + 2abc = 1$

99. Since, the given system of equations possess non-trivial solution, if

$\begin{vmatrix} 0 & 1 & -2 \\ 0 & -3 & 1 \\ k & -5 & 4 \end{vmatrix} = 0$

$\Rightarrow k = 0$

On solving the equations $x = y = z = \lambda$ (say)

$\therefore For\ k = 0,$ the system has infinite solutions of $\lambda \in R$.

100. for $\lambda \neq -5$ the system has atleast one solution

for $\lambda = -5$ the system has infinitly many solutions

$$x = \frac{4-5k}{7}, y = \frac{13k-9}{7}, z = k\,(k \in R)$$